

yubiOS Quarterly Business Review

Hardware-Anchored Sovereignty

FIDO2-First Immutable Linux



Project State and Engineering Velocity

Velocity

64+ PRs
Merged

14 in last 7 days



Stability

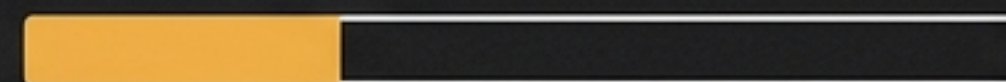
20 Green
Workflows



Phase

Phase 0:
Groundwork

Current



30%

Gate 1:
Technical Preview

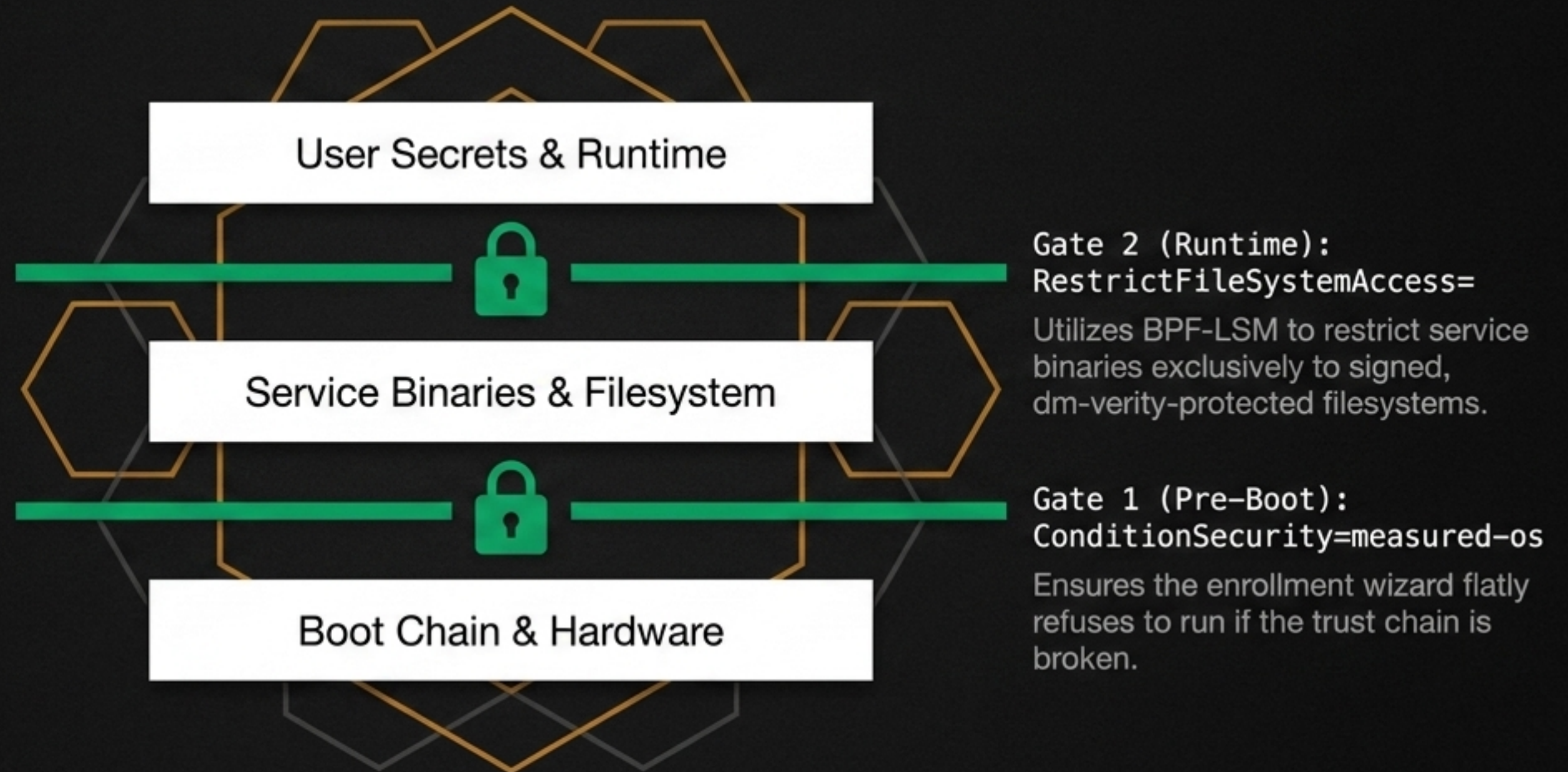
The project is rapidly executing its Phase 0 groundwork, successfully laying the foundation for a hardware-anchored trust chain.

The Sovereign Trust Anchor Shift

	Traditional Linux Security	yubiOS Architecture
Root of Trust	Platform TPM (OEM controlled)	YubiKey 5 Series (Owner controlled)
Secure Boot	Microsoft / Vendor Keys	Owner PIV Slot 9c via PKCS#11
Disk Unlock	TPM PCR Hashes (Brittle to updates)	FIDO2 hmac-secret (Update survivable)
OS Integrity	Mutable Packages	bootc OCI Image with dm-verity /usr

No OEM. No trust anchors you don't control.

Hardening the Runtime Baseline with systemd v261



yubiOS adopts these features to ensure the OS attack surface is reduced to the bare minimum before any user secrets are released.

Owning the Stack Below the Kernel

x86-64 remains fully supported above the UEFI layer, but ARM64 serves as the primary target (ADR-023) because it eliminates reliance on proprietary OEM firmware and discrete hardware TPMs.

**100%
Owner-
Provisioned
Code**

ARM64 Path A

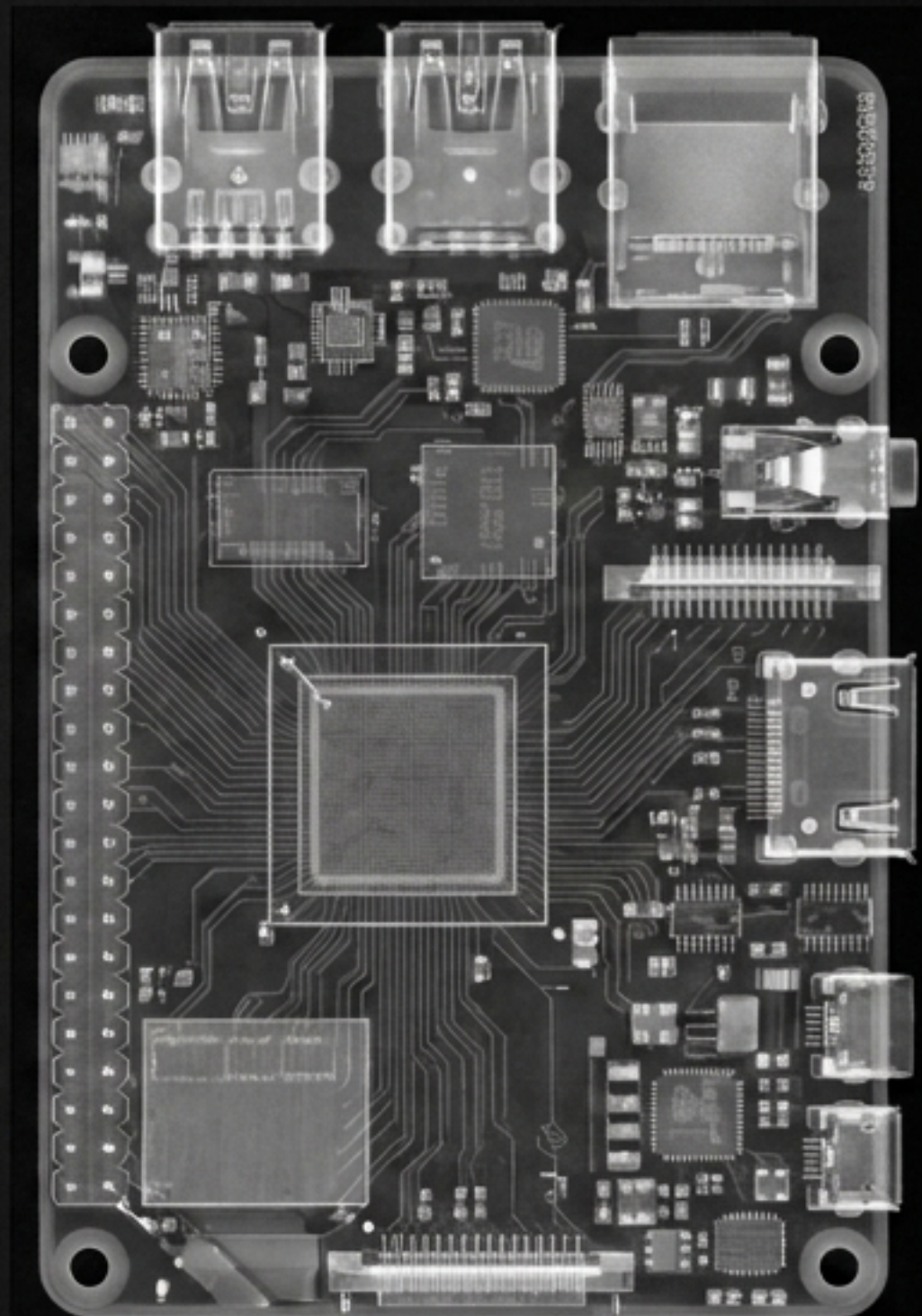
Boot ROM
(Owner ROTPK)

TF-A

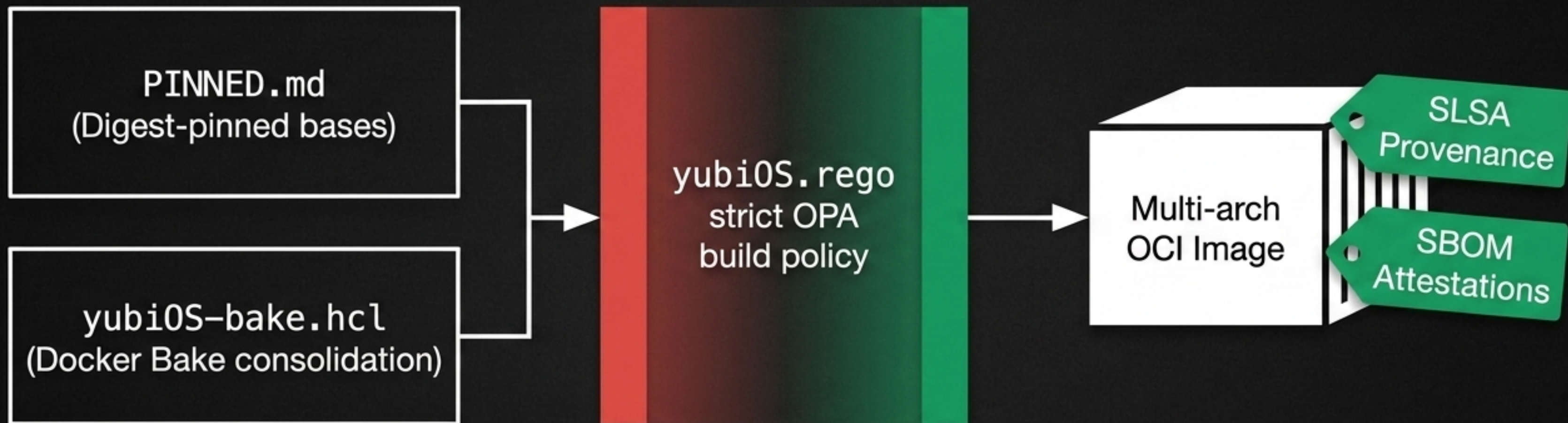
OP-TEE
(Hosting fTPM)

U-Boot

Signed UKI

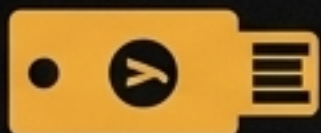


Cryptographically Verified Build Pipelines



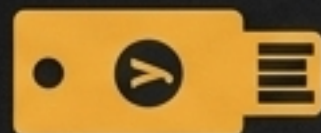
Key Takeaway: yubiOS is engineered to be AI-resilient; deployed authority relies on strict cryptographic verification and immutable digests, completely removing trust in the author—human or machine.

Zero to Sovereign in Four Steps



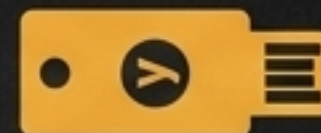
Step 1: PIV Secure Boot

```
Signing the UKI via  
PKCS#11...
```



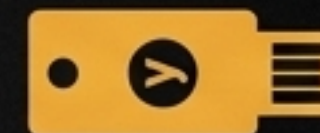
Step 2: LUKS2 hmac-secret

```
Binding the root  
filesystem...
```



Step 3: SSH Resident

```
Generating ed25519-sk  
keys...
```

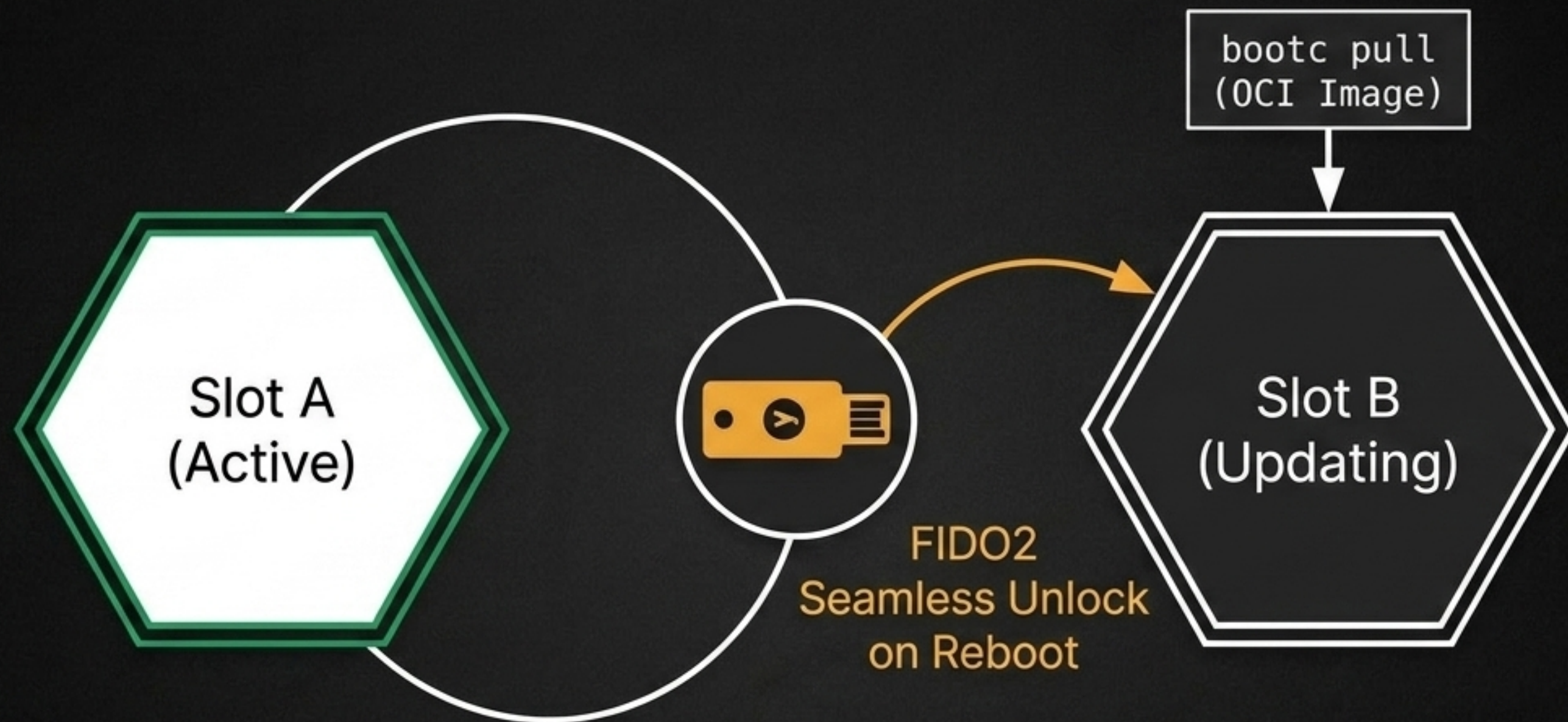


Step 4: PAM/U2F

```
Registering touch for  
sudo/login...
```

The `yubiOS-enroll` service automatically runs on `tty1` at first boot, anchoring every operational trust boundary to the physical token.

Survivable and Atomic System Updates



Because yubiOS uses FIDO2 hmac-secret for LUKS2 rather than brittle TPM PCR hashes, atomic OS updates occur without ever requiring the user to re-enroll their disk decryption secrets.

Defending the Faux Phy Attack Chain



Blocked by Design

- /usr bind-mount poison
(Blocked by dm-verity on every IO)
- Evil-twin EDK2
(Blocked by single Signed UKI)



Detected / Monitored

- Absolute Persistence/Computrace
(Detected via chipsec at first boot)
- OEM ROM supply chain
(Monitored, hardware out-of-scope)

Key Takeaway: yubiOS categorically eliminates pre-init hijacks and runtime control attacks by validating every component before execution.

The Critical Path to Unblocking

ID: B-ARM64-PATHA

Risk Area: Hardware

Resolution: Run documented sacrificial ROCK 5B rehearsal before production language.

ID: B-VM-CTAP2

Risk Area: CI

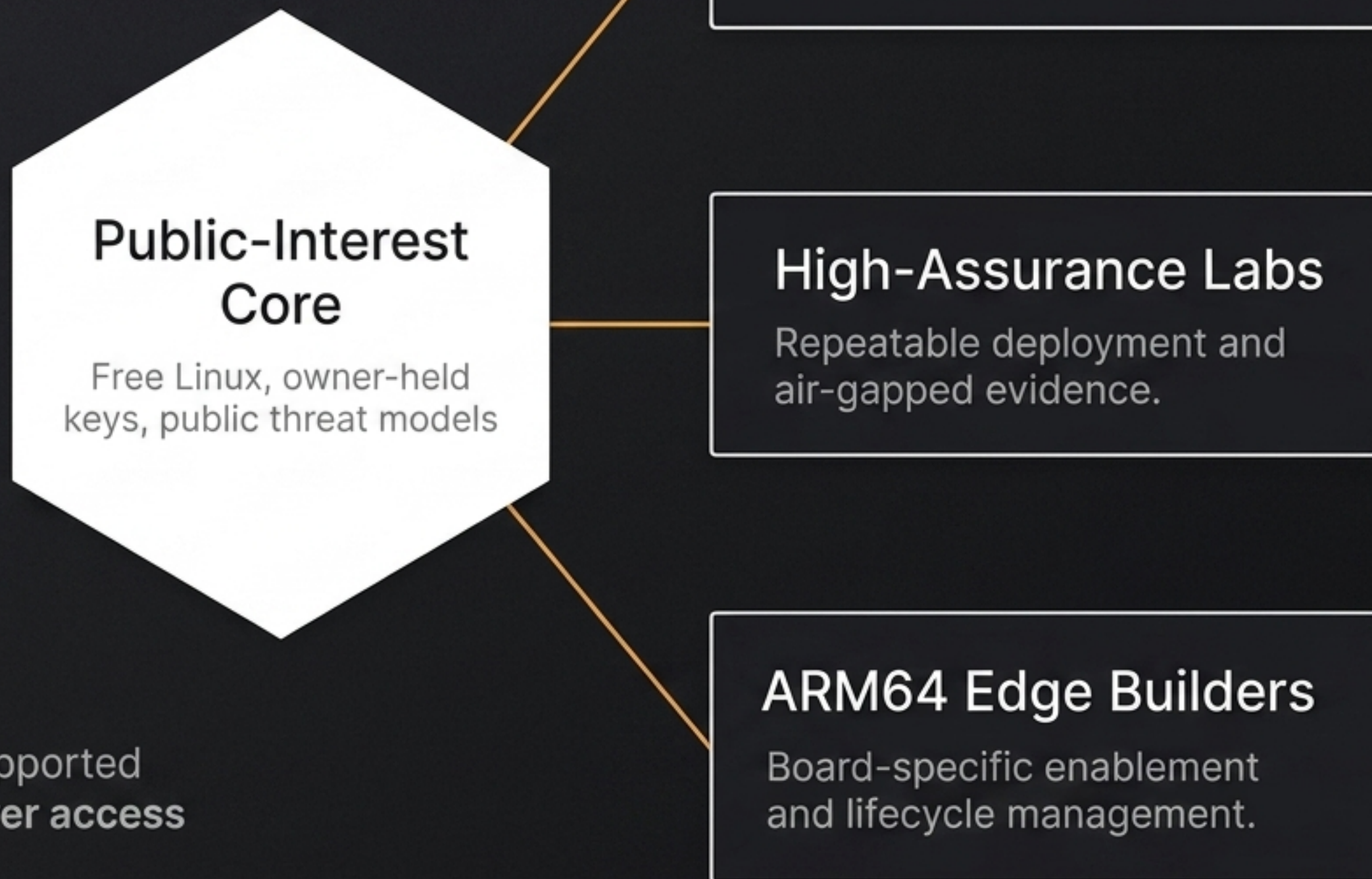
Resolution: Fix passless device path and assert token discovery before dependent tests.

ID: B-B00TC-SEAL

Risk Area: Supply Chain

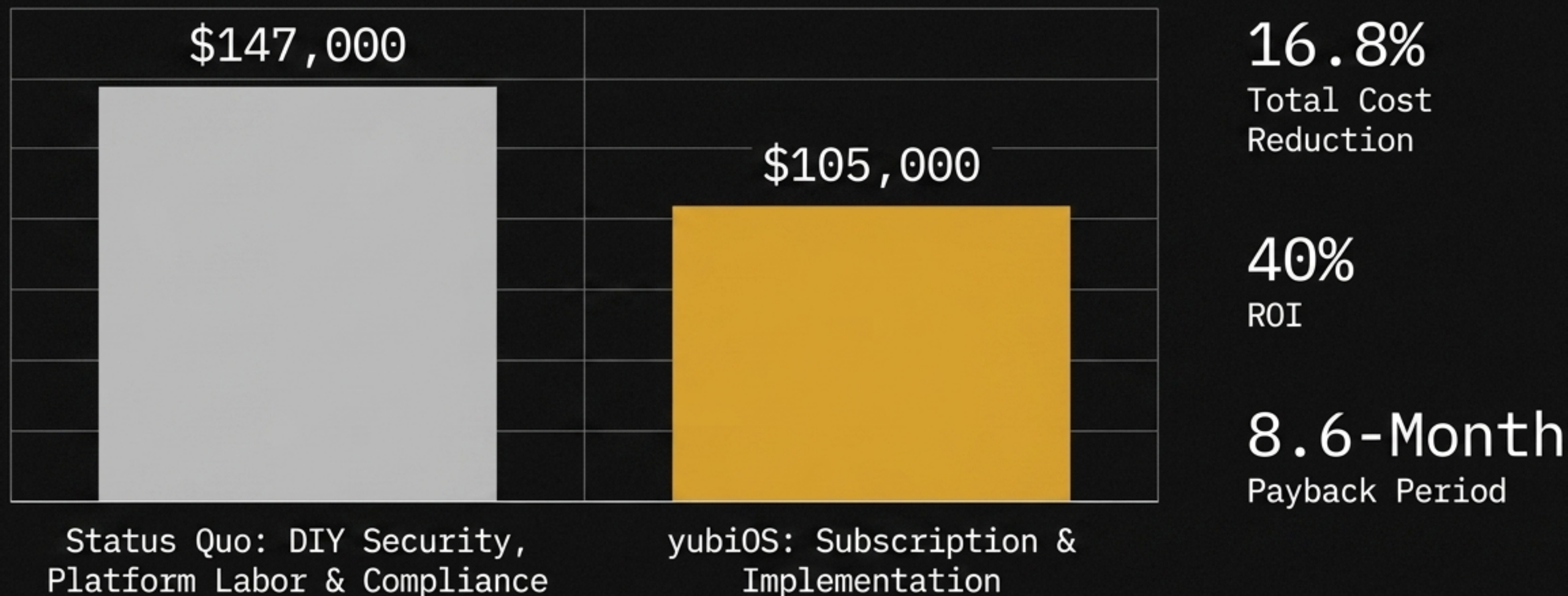
Resolution: Pin Fedora base with bootc v1.16.4 for split-kernel-and-rootfs capabilities.

Accountable Operations Over Code Paywalls



The commercial model sells integration, supported releases, and recovery accountability—**never access** to the **security-critical** source code itself.

The Assured Fleet Economic Model



The economic comparison is not "free Linux vs. paid Linux," it is yubiOS operations versus the vast internal cost of custom security engineering and compliance evidence assembly.

The Path to Technical Preview

Gate 0: Current Groundwork

Resolving VM/FIDO2 blockers, defining pilot platform, trademark clearance.

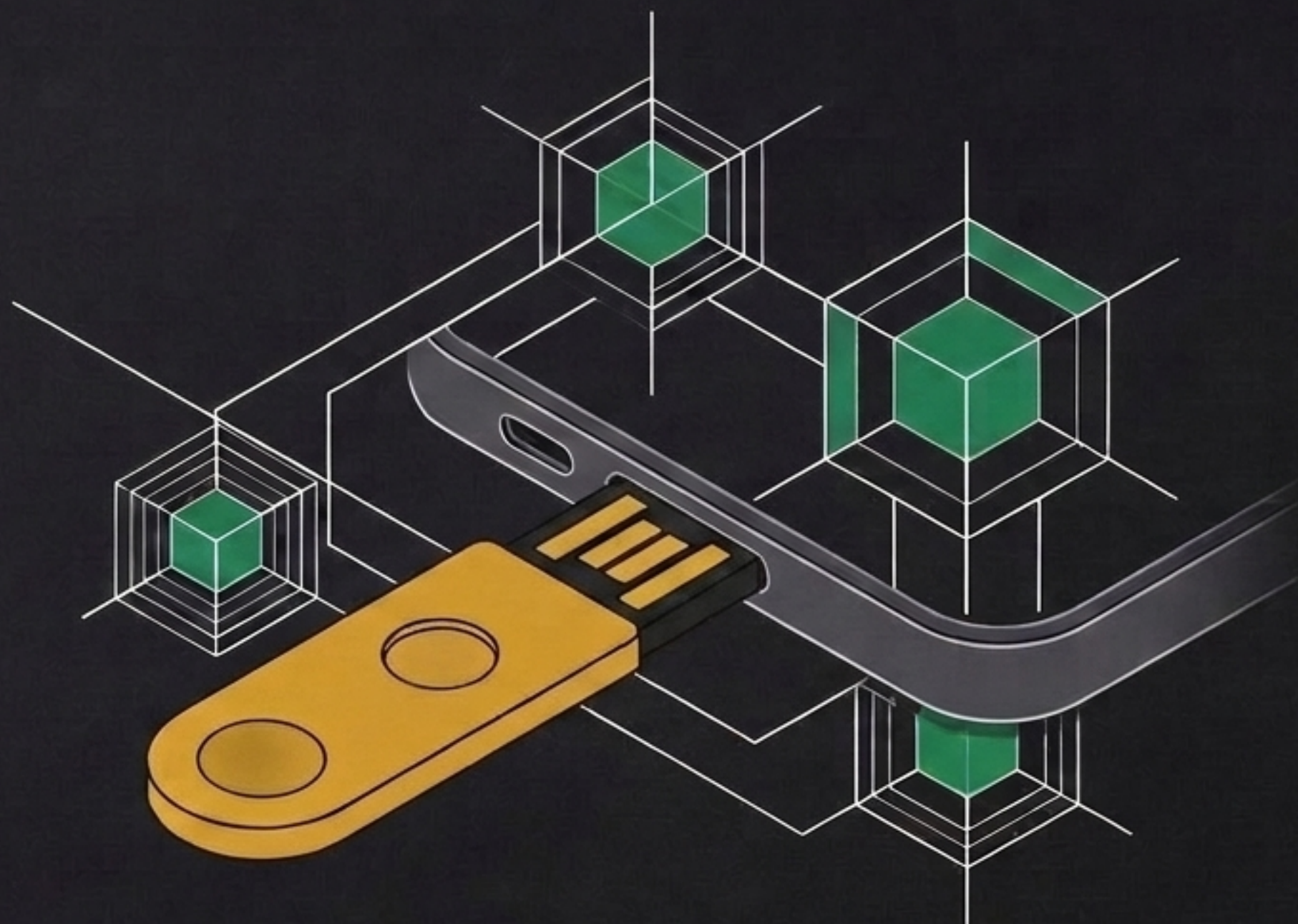
Gate 1: Technical Preview

Achieving repeatable physical-YubiKey demonstrations, signed verification material, published recovery exercises.

Gate 2: Proof Milestone

One key, four owner workflows, one reproducible proof on tested hardware configurations.





AI-Resilient Systems Built on Human Trust

yubiOS is built, reviewed, and tested with heavy AI assistance. The systems we ship must hold up even when the thing that built them cannot be trusted. The answer is structural: deployed authority depends on physical cryptographic verification, not trust in an author.